You are given an array of N integers representing the weights of people.

Your task is to divide the people into two groups such that:

The two groups have the same number of people, or their sizes differ by at most one. The absolute difference between the total weights of the two groups is minimized.

Print the minimum possible difference between the total weights of the two groups.

Input Format

The first line contains an integer N, the number of people. The second line contains N space-separated integers representing their weights.

Constraints

2 ≤ N ≤ 30 1 ≤ arr[i] ≤ 1000

Output Format

Print a single integer representing the minimum possible difference between the total weights of the two groups.

Sample Input 0

```
4
3 4 5 6
```

Sample Output 0

```
0
```

Sample Input 1

```
5
1 2 3 4 5
```

Sample Output 1

```
1
```

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```python
# Enter your code here. Read input from STDIN. P
n = int(input())
a = list(map(int, input().split()))

k = n // 2
total = sum(a)

left = a[:n//2]
right = a[n//2:]

L = [[] for _ in range(len(left) + 1)]
R = [[] for _ in range(len(right) + 1)]

for mask in range(1 << len(left)):
    s = 0
    cnt = 0

    for i in range(len(left)):
        if mask & (1 << i):
            s += left[i]
            cnt += 1

    L[cnt].append(s)

for mask in range(1 << len(right)):
```

Line: 61 Col: 11

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Test against custom input

Test case 0

Test case 1

Compiler Message

Success